

VOLUME I

Vision & Research

Project: APEX — Adaptive Personal Execution Engine

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Purpose: Establish the research foundation, vision, motivation, terminology, and market context for Project APEX.

"Learning Humans, Not Just Tasks."

Chapter 1 — Executive Summary

Project APEX proposes a general-purpose AI platform that learns professional workflows by observing how an individual works over time. Rather than relying only on prompts or predefined automations, the platform constructs a behavioral model that can assist, predict, and eventually automate repetitive workflows under human supervision.

Chapter 2 — Problem Statement

Current AI systems excel at answering questions and executing instructions, but they generally lack an understanding of an individual's workflow, preferences, and decision patterns. The core problem addressed by APEX is how to build an AI system that learns from observation while respecting privacy and maintaining human control.

Chapter 3 — Vision & Mission

Vision: Build AI systems that understand how people work. Mission: Create a reusable platform capable of observing workflows, extracting reusable skills, modeling decision-making, and collaborating safely with users across domains.

Chapter 4 — Core Research Questions

How should human behavior be represented? Which events are worth observing? How can skills be extracted from workflows? How should confidence and autonomy increase over time? How can privacy and explainability remain first-class principles?

Chapter 5 — Research Methodology

The project follows a research-first methodology: literature review, architecture design, observation engine, workflow mining, behavioral modeling, prototype validation, and iterative evaluation with real users. The first validation domain will be a professional wedding album designer.

Chapter 6 — Deliverables

This volume establishes terminology, defines the project scope, identifies research gaps, and prepares the foundation for the remaining architecture volumes.

Volume I Roadmap

- Executive Summary
- Problem Statement
- Vision & Mission
- Literature Review
- Market Analysis
- Research Gaps
- Terminology
- Success Metrics
- Scope & Non-Scope
- Transition to Volume II

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